Diskrete Mathematik

Woche 6 - HG G 26.5

https://github.com/tschai-yim/eth-resources

Feedback (anonym): https://forms.gle/xnkcjsdj86LK6yMh7

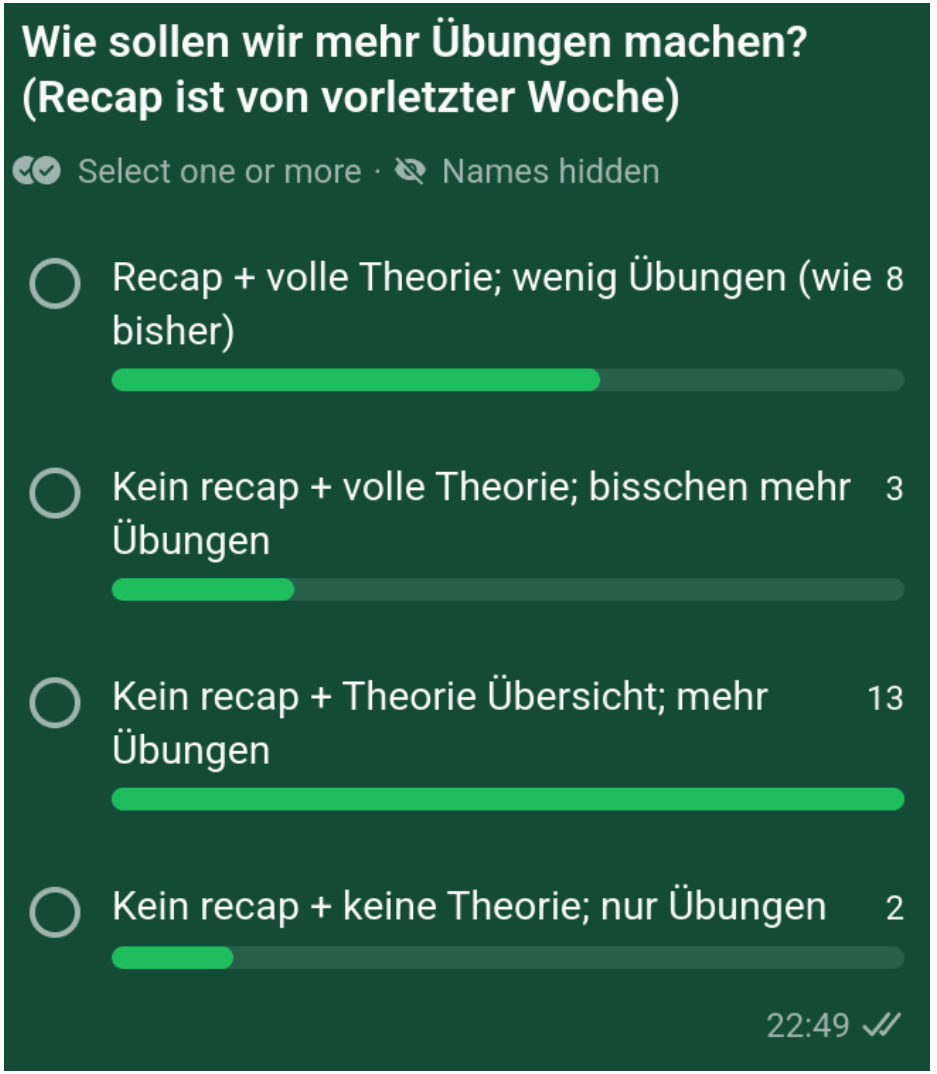


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Resultat der Umfrage

- Durchschnitt ≈ 1.34 von $0 - 3$
- **Änderungen:**
  - Kein recap
  - Theorie weniger detailliert
  - Unwichtiges wird geskipped
    z.B. Hasse Diagrams, Lattices
- **Wer mehr Theorie will:**
  - Zusammenfassungen erstellen
  - *pot.* meine in `eth-resources` lesen
  - *pot.* Skript detailliert lesen



Heutige Agenda

1. Teil (45 min)
- 20 min: Theorie-Übersicht mit Beispielen
- 25 min: Übungen

10 min Pause (hier persönliche Fragen)

2. Teil (45 min): Übungen

Theorie-Übersicht

Kapitel 3.6-3.7

Relations: Operations (recap)

Forgot to show calculation trick:

- **Transitive Closure (ρ^*):**
  - Union of all powers: $\rho^* = \bigcup_{n \in \mathbb{N} \setminus \{0\}} \rho^n = \rho \cup \rho^2 \cup \rho^3 \cup \ldots$. [Def 3.18]
  - Represents **reachability**: $a \rho^* b \Leftrightarrow$ path of finite length exists from a to b.
  - Example: $\{(a, b), (a, c), (b, a), (c, b)\}$

Functions

- **Definition ($f : A \to B$):** Relation from domain A to codomain B that is:
  - **Total:** every $a \in A$ is mapped. (**Partial function**: not necessarily total)
  - **Well-defined:** each $a \in A$ maps to a **unique** $b \in B$.
- **Types:**
  - **Injective**: $f(a_1) = f(a_2) \Rightarrow a_1 = a_2$ (1-to-1 mapping).
  - **Surjective**: $\mathrm{im}(f) = B$ (every element in codomain reached).
  - **Bijective:** both injective and surjective.
- **Operations:**
  - **Composition:** $(g \circ f)(a) = g(f(a))$ (**opposite order** to relations).
    - Properties like injectivity/surjectivity are preserved.
  - **Inverse ($f^{-1} : B \to A$):** exists $\Leftrightarrow$ f is **bijective**.
  - **Preimage ($f^{-1}(T)$):** defined for any $T \subseteq B$, even *without an inverse function.*

Countability

- **Comparing Set Sizes:**
  - **Equinumerous ($A \sim B$):** bijection exists (same size). [Def 3.42]
  - **Domination ($A \preceq B$):** injection $A \to B$ exists (A "no larger" than B).
  - **Bernstein-Schröder:** $A \preceq B \land B \preceq A \Rightarrow A \sim B$. [Thm 3.16]
- **Countable ($A \preceq \mathbb{N}$):** iff A is finite or $A \sim \mathbb{N}$ (no infinity in between). [Thm 3.17]
  - **Examples:** $\mathbb{Z}, \mathbb{Q}, \mathbb{N} \times \mathbb{N}$, and finite bitstrings $\{0,1\}^*$.
- **Uncountable:** not countable, e.g., infinite sequences $\{0,1\}^\infty \sim P(\mathbb{N})$.
  - **Cantor's Diagonalization:**
    1. assume list of all infinite sequences exists
    2. flip bits on the diagonal
    3. yields sequence not in list (contradiction).

Übungen

https://github.com/tschai-yim/eth-resources unter
`03_TA/26HS_252-0025-01L_Diskrete_Mathematik/exercises`

Problem 1: Functions as Relations & Function Sets

Def 3.33 Function $f : A \to B$: relation $f \subseteq A \times B$ with:

1. **totally defined**: $\forall a \in A \; \exists b \in B : (a, b) \in f$
2. **well-defined**: $(a, b) \in f \wedge (a, b') \in f \Rightarrow b = b'$

Def 3.35 Partial function: only condition (2) (well-defined).

Def 3.9 Relation from A to B: any subset $\rho \subseteq A \times B$.

Def 3.34 Set B^A: all functions $A \to B$. Finite: $|B^A| = |B|^{|A|}$.

Problem 2: Images Under Functions & Intersections

Def 3.36 Image: $f(S) := \{f(a) \mid a \in S\} = \{b \in B \mid \exists a \in S : f(a) = b\}$.

Def 3.39 Injective: $a \neq a' \Rightarrow f(a) \neq f(a')$.
Contrapositive: $f(a) = f(a') \Rightarrow a = a'$.

Def 3.2 / Lem. 3.2 Subset & Set equality:
- $X \subseteq Y \Leftrightarrow \forall x(x \in X \Rightarrow x \in Y)$
- $X = Y \Leftrightarrow X \subseteq Y \land Y \subseteq X$

Problem 3: Composition of Functions & Relations

Def 3.41 Function composition & Relation order:
$(g \circ f)(a) := g(f(a))$. As a relation: $g \circ f = f \circ g$.

Def 3.12 Relation composition: $\rho \circ \sigma := \{(a, c) \mid \exists b : (a, b) \in \rho \wedge (b, c) \in \sigma\}$.

Def 3.39 For $f : A \to B$:

- **Injective**: $f(a) = f(a') \Rightarrow a = a'$
- **Surjective**: $\forall b \in B \; \exists a \in A : f(a) = b$ (i.e. $f(A) = B$)
- **Bijective**: both injective and surjective

Problem 4: Equinumerosity & Proper Subsets

Def 3.42 (i) Equinumerous: $A \sim B \Leftrightarrow \exists$ bijection $f : A \to B$.

Def 3.39 Bijection: both injective and surjective.

Lem 3.15 (i) Equivalence relation: $\sim$ is reflexive, symmetric, and transitive ($A \sim B \wedge B \sim C \Rightarrow A \sim C$).

Ref §3.7.2 Proper subsets: A finite set is never equinumerous to a proper subset.

Problem 5: Countable and Uncountable Sets

Def 3.42
- $A \preceq B \Leftrightarrow \exists$ injection $f : A \to B$.
- A **countable** $:\Leftrightarrow A \preceq \mathbb{N}$.

Cor 3.20, 3.21 & Ex. 3.56
- $\mathbb{Z}$ and $\mathbb{Q}$ are countable ($\mathbb{Z} \sim \mathbb{N}, \mathbb{Q} \preceq \mathbb{N}$).
- A, B countable $\Rightarrow A \times B$ countable.

Lem 3.15 (ii) Transitivity: $A \preceq B \wedge B \preceq C \Rightarrow A \preceq C$.

Thr 3.22 (iii) A countable $\Rightarrow A^*$ (finite sequences) countable.

Thr 3.23 $\{0,1\}^\infty$ is uncountable.

Problem 6: Uncountability via Injection from $\{0,1\}^\infty$

Def 3.42 & Lem. 3.15 (ii) Dominance & Transitivity:
- $A \preceq B \Leftrightarrow \exists$ injection $f : A \to B$.
- $A \preceq B \wedge B \preceq C \Rightarrow A \preceq C$.
- A countable $\Leftrightarrow A \preceq \mathbb{N}$.

Thr 3.23 $\{0,1\}^\infty$ is uncountable.

Def 3.33 Function: $f : A \to B$ requires $\forall a \in A \; \exists b \in B : (a,b) \in f$.

Def 3.39 Injective: $a \neq a' \Rightarrow f(a) \neq f(a')$.

Problem 7: Surjective Images of Countable Sets

Def 3.42 (iii) B countable $\Leftrightarrow \exists$ injection $g : B \to \mathbb{N}$.

Def 3.38 & 3.39 Preimage & Surjectivity:

- Preimage: $f^{-1}(\{b\}) := \{a \in A \mid f(a) = b\}$.
- Surjective: $\forall b \in B\ \exists a \in A : f(a) = b$ (i.e. $f^{-1}(\{b\}) \neq \varnothing$).

Thr 3.17 / Def. 3.30 Well-ordering of $\mathbb{N}$: Every non-empty subset $S \subseteq \mathbb{N}$ has a least element $\min(S) \in S$.

Lem 3.15 (ii) Transitivity: $A \preceq B \wedge B \preceq C \Rightarrow A \preceq C$.

Problem 8: Cantor's Theorem & Diagonal Argument

Def 3.7 & 3.42 Power Set & Dominance:
- $\mathcal{P}(A) := \{S \mid S \subseteq A\}$.
- $A \preceq B \Leftrightarrow \exists$ injection $A \to B$.

Thr 3.17 Countable set: S is countable $\Leftrightarrow$ S is finite or $S \sim \mathbb{N}$.

Thr 3.23 Cantor's diagonalization: For any sequence $(\beta_i)_{i \in \mathbb{N}}$, the sequence α with $\alpha_i \neq \beta_{i,i}$ satisfies $\forall k : \alpha \neq \beta_k$.

Problem 9 (extra): Injective ⇔ Surjective for Finite Sets

Def 3.39 For $f : A \to B$:
- **Injective**: $f(a) = f(a') \Rightarrow a = a'$
- **Surjective**: $f(A) = B$

Def 3.36 & 3.37 Image & Cardinality: For finite A: $|f(A)| \leq |A|$, with equality ⇔ f is injective.

Ref §3.7.2 Finite subsets: For finite B and $S \subseteq B$:

$$|S| = |B| \Leftrightarrow S = B$$

Problem 10 (extra): Disjoint Intervals in $\mathbb{R}$

Def 3.42 $\mathcal{J}$ countable $\Leftrightarrow \exists$ injection $\mathcal{J} \to \mathbb{N}$.

Cor 3.21 $\mathbb{Q}$ is countable ($\exists$ injection $q : \mathbb{Q} \to \mathbb{N}$).

Thr 3.17 / Def. 3.30 Well-ordering of $\mathbb{N}$: Every non-empty subset $S \subseteq \mathbb{N}$ has a least element $\min(S) \in S$.

Ref §3.7.4 Uncountability of $\mathbb{R}$: $\mathbb{R}$ is uncountable ($\mathbb{R} \sim \{0,1\}^{\infty}$).

Offene Fragen & Feedback

- Gibt es noch Fragen zu den heutigen Themen oder den Übungen?
- Habt ihr Feedback zur heutigen Übungsstunde?
  - Anonymes Formular: https://forms.gle/xnkcjsdj86LK6yMh7
- Bei Fragen: Schreibt mir eine E-Mail oder auf WhatsApp.

Schöne Woche und bis nächsten Montag!